Ang Cao

Research Scientist, Google DeepMind

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Research Interests

My research aims to advance the capabilities of frontier AI models. I am particularly interested in discovering scalable, self-sustaining sources of learning signals that enable these models to continually improve beyond the limits of curated human supervision. This includes agentic discovery loops for continual model improvement.

Research Appointments

Research Scientist, *Google DeepMind, Mountain View* Dec. 2025–Present

Developing multimodal and agentic capabilities for Gemini, focusing on multimodal coding and autonomous research to improve large-scale pretraining.

Projects as a core contributor or lead:

- Multimodal/visual agentic coding for Gemini.
- Autonomous research for large-scale pretraining — Gemini recursive self-improvement (RSI).
- Gemini-Omni 3D/4D world modeling and agentic environments for robotics.

Research Scientist Intern, *Meta FAIR, Menlo Park* May 2024–Dec. 2024

Mentors Sasha Sax and Franziska Meier.

Focus Proposed LIFT-GS, a highly data-efficient 3D vision-language grounding (VLG) pipeline that learns from foundation VLMs through differentiable rendering, without requiring 3D annotations.

Research Scientist Intern, *Meta GenAI, London* May 2023–Nov. 2023

Mentors David Novotny, Andrea Vedaldi, and Natalia Neverova.

Focus Designed highly efficient GPU kernels for neural rendering (Lightplane), achieving up to a 10,000× reduction in rendering memory usage for feed-forward 3D foundation models.

Education

Ph.D., Computer Science and Engineering, *University of Michigan* 2020–2025 | GPA 4.0/4.0

Thesis: *Toward Spatial Intelligence via Data and Compute Efficiency*. Advisor: Justin Johnson.
Committee: Andrew Owens, Jeong Joon Park, Andrea Vedaldi, and Stella Yu.

M.S., Electrical and Computer Engineering, *University of Michigan* 2018–2020 | GPA 4.0/4.0

B.S., Electrical Engineering, *Wuhan University* 2014–2018 | GPA 3.9/4.0

Manuscripts in Preparation

"Autonomous Pretraining at Scale"

"CE Bench: Can Agents Predict What a Training Run Will Do?"

"The Flip Test: Do World Models Change Their Minds When the Physics Does?"

Publications

* Equal contribution. † Mentored project. Citation record on Google Scholar.

[1] **Generative Point Tracking and Forecasting**
Xuanchen Lu, **Ang Cao**†, Chao Feng, Andrew Owens
CVPR 2026 | Project

[2] **Probing Visual Language Priors in VLMs**
Tiange Luo*, **Ang Cao***, Gunhee Lee, Justin Johnson, Honglak Lee
ICML 2025 | Project

[3] **From Thousands to Billions: 3D Visual Language Grounding via Render-Supervised Distillation from 2D VLMs**
Ang Cao, Sergio Arnaud, Oleksandr Maksymets, …, Alexander Sax
ICML 2025 | Project

[4] **LOCATE 3D: Real-World Object Localization via Self-Supervised Learning in 3D**
Paul McVay, Sergio Arnaud, Ada Martin, …, **Ang Cao**, …, Franziska Meier
ICML 2025 | Project

[5] **Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass**
Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, **Ang Cao**, Joyce Chai, Franziska Meier, Matt Feiszli
CVPR 2025 | Project Code

[6] **Lightplane: Highly-Scalable Components for Neural 3D Fields**
Ang Cao, Justin Johnson, Andrea Vedaldi, David Novotny
3DV 2025 | Project Code

[7] **Meta 3D Gen**
Raphael Bensadoun, Tom Monnier, Yanir Kleiman, …, **Ang Cao**, …, Andrea Vedaldi
Technical report, 2024 | Project

[8] **DreamGaussian4D: Generative 4D Gaussian Splatting**
Jiawei Ren*, Liang Pan*, Jiaxiang Tang, Chi Zhang, **Ang Cao**, Gang Zeng, Ziwei Liu
arXiv 2023 | Project

[9] **Text2Room: Extracting Textured 3D Meshes from 2D Text-to-Image Models**
Lukas Höllein*, **Ang Cao***, Andrew Owens, Justin Johnson, Matthias Nießner
ICCV 2023 Oral | Project Code

[10] **HexPlane: A Fast Representation for Dynamic Scenes**
Ang Cao, Justin Johnson
CVPR 2023 | Project Code

[11] **FWD: Real-Time Novel View Synthesis with Forward Warping and Depth**
Ang Cao, Chris Rockwell, Justin Johnson
CVPR 2022 | Project Code

[12] **Inverting and Understanding Object Detectors**
Ang Cao, Justin Johnson
Technical report, 2021 | Paper Code

[13] **Unified Signal Compression Using Generative Adversarial Networks**
Bowen Liu*, **Ang Cao***, Hun-Seok Kim
ICASSP 2020 Oral | Paper

Honors and Awards

Rollin M. Gerstacker Foundation Fellowship, University of Michigan 2020
Outstanding Graduate of Wuhan University 2018
National Scholarship, Ministry of Education of China 2016, 2017
Meritorious Winner, Mathematical Contest in Modeling 2017
National Undergraduate Innovation Foundation, Ministry of Education of China 2016

Professional Service

Outstanding Reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026

Conference reviewer: CVPR (since 2022), NeurIPS (since 2023), ICLR (since 2023), ECCV (since 2022), ICCV (since 2023), SIGGRAPH, and SIGGRAPH Asia.

Journal reviewer: IEEE TPAMI and IEEE TVCG.

Invited Talks

"Toward Building a Realistic Digital 3D World." NVIDIA Spatial Intelligence Lab; Meta FAIR; Google DeepMind; Meta Reality Labs; Stanford Vision and Learning Lab 2025